

Anil Praneeth Vure

Solutions Architecture | Applied AI | Cloud Deployment

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Solutions architect with 20+ years of experience, including AWS, Informatica, and Wipro. Combine customer-facing technical leadership with applied AI and generative AI engineering across manufacturing analytics, agent-assisted software development, and CAD/simulation.

Technical focus: AI agents and evaluation, RAG/GraphRAG, AWS deployment, Python/C++/CUDA, industrial IoT and simulation.

Professional Experience

Amazon Web Services

Feb 2019 - Present

Senior Data Architect / Data Architect, IoT Analytics | Austin, TX

- Built global overall equipment effectiveness (OEE) analytics for an automotive manufacturer, enabling shift and shop-floor comparisons that helped improve productivity and reduce cycle times.
- Translate manufacturing requirements into AWS architectures for industrial IoT (IIoT), machine learning, computer vision, time-series analytics, and edge-to-cloud integration.
- Partner with sales, business development, and technical teams on workload strategy and AWS adoption; lead architecture discussions and create executive-ready solution guidance and field enablement.

Selected AI Engineering Projects

Quatrion

Manufacturing agent platform in development | github.com/quatrion

- Directed coding agents to refactor a Python FEA solver into C++/CUDA for NVIDIA GPUs, using knowledge-graph grounding and multi-agent debate to guide implementation.
- Implemented spatial tokenization and bounded knowledge graphs linking CAD parts, assembly hierarchy, 3D positions, orientations, and surface-distance relationships to ground LLM context.
- Built nonlinear FEA solvers with FEniCSx/PETSc for large deformation and J2 plasticity, with analytical benchmarks; developed CFD validation/postprocessing and OpenRadioss sheet-metal forming workflows.
- Created reusable engineering patterns and agent skills for workflow automation; automated Kubernetes/Helm deployments with authentication gates and CI checks for cloud and on-premises targets.

Personalized Learning App

Agent-assisted development for two learners | Project overview

- Built a React/TypeScript application with Codex agents: four reading and mathematics courses, 48 chapters, interactive lessons, and parent review workflows.
- Integrated Bedrock tutoring and Polly narration with strict-schema response validation, regression tests for malformed or lesson-incompatible outputs, and fallback guidance; isolated learner records and protected server credentials.
- Applied an AI-assisted software development lifecycle (SDLC) with parallel agents for implementation and review; a September 2026 run passed 682 tests, lint, and a production build, alongside browser validation.

Technical Skills

AI agents: Amazon Bedrock, Strands Agents, LangChain; multi-agent orchestration and debate, agent memory, planning, failure recovery, context engineering.

Retrieval and integration: Retrieval-augmented generation (RAG), GraphRAG, knowledge-graph grounding, tool calling, Model Context Protocol (MCP) server development, LLM application integration.

Agent evaluation: Evaluation harnesses, regression evaluation, response validation; familiarity with Petri/Bloom-style behavioral evaluations, multi-turn scenarios, and model-based grading.

Model lifecycle: Model serving, inference optimization, model customization, fine-tuning, LoRA and parameter-efficient fine-tuning (PEFT).

Cloud and deployment: AWS, Amazon Bedrock AgentCore, Docker; container orchestration (Kubernetes/Helm), infrastructure as code (IaC, CloudFormation), CI/CD, observability.

ML and data: Broad familiarity across the AWS ML and data analytics stack; computer vision, time series, Spark, Kafka, PostgreSQL, ETL/ELT, data governance.

Software engineering: Python, TypeScript, React, SQL; agent-assisted C++/CUDA refactoring, Codex/Claude Code workflows, automated testing and browser validation.

CAD and simulation: Spatial tokenization, nonlinear FEA, FEniCSx/PETSc, NVIDIA CUDA, CFD validation/postprocessing, OpenRadioss.

Technical leadership: Executive advisory, mentoring, technical roadmap ownership, organization-wide adoption, customer enablement.

Earlier Professional Experience

Informatica

Mar 2012 - Jan 2019

Big Data Specialist / Solutions Architect | Redwood City, CA

- Led customer-facing architecture and enterprise delivery for data lakes, governance, MDM, and ETL/ELT, including implementation planning and operational readiness for go-live.
- Built batch and streaming pipelines; translated customer requirements into product feedback and published knowledge-base and Velocity articles for field enablement.

Wipro

Mar 2008 - Mar 2012

Senior Software Engineer - Data Services | Bangalore, India

- Delivered data warehousing, migration, MDM, and data quality projects using Informatica; implemented ETL performance tuning, error handling, and recovery procedures.

Additional Developer Tooling

Coding Best Practices MCP Server

github.com/anilpraneeth/coding-best-practices-mcp-server

- Built Python/AWS developer tooling through an MCP server for linting, security analysis, infrastructure checks, cost optimization, and project standards.

Education and Credentials

B.Eng., Electronics and Communication: Sri Venkateshwara University, India | 2007

AWS certifications: Machine Learning Engineer - Associate (2025); AI Practitioner (2025).

Awards: AWSome Awards All-Star Recipient (2025); Informatica Customer Dedication Award and Spotlight Award.